



NAME:
HABIB ID:

SECTION NUMBER:
INSTRUCTOR:

LINEAR ALGEBRA, MIDTERM [Total Marks: 100]

(SPRING 2022)

Question 1: [10 Marks]

Prove that for an invertible matrix B , its inverse is unique. [6 MARKS]

For a matrix C , if we know that $\det(C) > 0$, then show that $(C^{-1})^T = (C^T)^{-1}$. [4 MARKS]



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Question 2: [10 Marks]

Let Y and Z be $n \times n$ matrices.

(a) Prove that if $I_n = YZ$ then $Z = Y^{-1}$. [5 MARKS]

(b) Prove that if $I_n = ZY$ then $Z = Y^{-1}$. [5 MARKS]



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Question 3 [10 MARKS]

If B is an $m \times m$ square matrix, and the homogenous system $B\underline{x} = \underline{0}$ has only the trivial solution, then show that the reduced row echelon form of B is the identity matrix I_m . [7 MARKS]

Give two different row Echelon forms for the matrix $\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$. [3 MARKS]



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Question 4 [10 MARKS]

Let $A\underline{x} = \underline{b}$ be any consistent system of linear equations and let $\underline{x}_1$ be a fixed solution. Show that every solution to the system can be written in the form $\underline{x} = \underline{x}_1 + \underline{x}_0$ where $\underline{x}_0$ is a solution of the homogenous linear system $A\underline{x} = \underline{0}$. Show also that every matrix of this form is a solution.

Question 5 [10 MARKS]

If $\underline{u}$ and $\underline{v}$ are vectors in $\mathbb{R}^n$ then prove that $\|\underline{u} + \underline{v}\| \leq \|\underline{u}\| + \|\underline{v}\|$. [5 MARKS]



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If $\underline{u}$, $\underline{v}$, and $\underline{w}$ are vectors in $\mathbb{R}^n$ then show that $d(\underline{u}, \underline{v}) \leq d(\underline{u}, \underline{w}) + d(\underline{w}, \underline{v})$. [5 MARKS]

Question 6 [10 MARKS]

Prove or disprove the following claim: “*It is possible for a vector space to contain only two distinct elements.*” [5 MARKS]

Let S be the set of all real numbers in $(0, \infty)$. If S is a vector space under the operations

$\underline{u} + \underline{v} = \underline{u} \underline{v}$ AND $k\underline{u} = \underline{u}^k$, then identify the zero element of S and show that it truly is the zero element. [5 MARKS]



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Question 7 [10 MARKS]

Prove that for any vectors $\underline{a}$, $\underline{b}$, and $\underline{c}$, the vectors $(\underline{a} - \underline{b})$, $(\underline{b} - \underline{c})$ and $(\underline{c} - \underline{a})$ form a linearly dependent set. [5 MARKS]

Prove that if $\{\underline{a}, \underline{b}\}$ is a linearly independent set, and $\underline{c} \notin \text{span}\{\underline{a}, \underline{b}\}$, then $\{\underline{a}, \underline{b}, \underline{c}\}$ is also linearly independent. [5 MARKS]



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Question 8 [10 MARKS]

Given a set S with two or more vectors, prove that if S is linearly independent then no vector in S is expressible as a linear combination of other vectors in S . [5 MARKS]

Given a set S with two or more vectors, prove that if no vector in S is expressible as a linear combination of other vectors in S , then S is linearly independent. [5 MARKS]



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Question 9 [10 MARKS]

Let S be a finite set of vectors in a ***finite dimensional*** vector space V . Then prove that if $\text{span}(S) = V$ but S is linearly dependent, then S contains a proper subset B (i.e. $B \subset S$) such that B is a basis for V .

Question 10 [10 MARKS]

Show that the set of all polynomials in P_n that have a horizontal tangent at $x = 1$ is a subspace of P_n .



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ROUGH WORK